MILO-weighted L1: from raw mask to bounded weights

balintio/milo-loss · MILOLoss

$$\mathcal{L} = \text{mean}\left(\text{sg}[\tilde{m}] \odot |\hat{y} - y|\right), \qquad \tilde{m} = \exp\left(\tfrac{1}{2} \ln R \cdot \tanh(z \ln S / \ln R)\right), \qquad z = \left(\log(m + \varepsilon) - \mu_b\right)/(\sigma_b + \varepsilon)$$

1 MILO mask



2 Log

$\ell = \log(m + \varepsilon)$

ε guards $\log 0$

3 Standardize

$z = \dfrac{\ell - \mu_b}{\sigma_b + \varepsilon}$

per image b, over all pixels

4 Soft-clip

$c \tanh(z/c)$

$c = \ln R / \ln S$: knee at $|z| \approx c$

5 Rescale, exp

$\tilde{m} = e^{\frac{r}{2} \tanh(z/c)}$

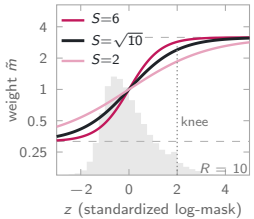
$r = \ln R$: $\tilde{m} \in (R^{-1/2}, R^{1/2})$

6 Weighted L1

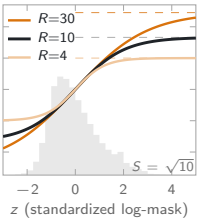
mean $(\tilde{m}\,|\hat{y} - y|)$

$\tilde{m}$ detached, shared by RGB

S sets the slope



R sets the ceiling



Black: defaults. Gray: z of real MILO masks (3 images × 5 distortions), right-skewed, so the $\sqrt{R}$ ceiling binds for the top few %.

S `target_weight_ratio` strength — default $e^{1.15} \approx 3.16$

Typical contrast: weights at the 90th and 10th percentile differ by $\approx S\times$.

▲ **Raise** (to ∼5) if the regions MILO flags still show the most visible errors.

▼ **Lower** (to ∼2, toward L1) if regions MILO deems unimportant lag behind.

R `max_weight_ratio` outlier cap — default $e^{2.3} = S^2 \approx 10$

Hard cap: weights stay in $(1/\sqrt{R}, \sqrt{R})$. With $R = S^c$ the cap engages at $|z| \approx c$; the default $c = 2$ caps only the top ∼4% of pixels.

▲ **Raise** ($R = S^3$: 0.5% capped) if many pixels pile up at $\sqrt{R}$.

▼ **Lower** ($R = S^{1.5}$: 12% capped) if a few hot spots dominate the loss.

ε eps $= 10^{-4}$: floors $\log(m + \varepsilon)$ and $\sigma_b + \varepsilon$; it barely acts, so leave it.

Tuning recipe: 1 pick S for strength (2 mild, 3.16 default, 5 strong) 2 set $R = S^2$ 3 move R only if the cap misbehaves.

Mean weight (≈ loss scale): 1.03 / 1.10 / 1.22 at $S = 2$ / 3.16 / 5, so retune the LR after big moves; inspect `normalized_mask()` early on.